HASSAM ASLAM

Unreal Engine Developer · C++ / Blueprints

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SUMMARY

Unreal Engine developer working in UE 5.8 C++ and Blueprints: combat, the Gameplay Ability System, enemy and NPC AI, multiplayer and UI. Shipped 12 Fortnite islands as Zeritress (855M+ minutes played). I build systems designers can tune through data, and I prove they work with automation tests and measured results.

SKILLS

Engine & code	Unreal Engine 5 (to 5.8), C++, Blueprints, Verse (UEFN), Visual Studio, Git
Gameplay	Gameplay Ability System (abilities, effects, attribute sets, tags, cues, prediction), combat, input buffering, cameras, Enhanced Input, inventory and crafting
AI	AI Controllers, behavior trees with custom C++ nodes, EQS, AI Perception, NavMesh and Detour crowd, LLM integration and retrieval
Multiplayer	Server-authoritative replication, client prediction, lag compensation, fast-array replication, listen and dedicated servers
UI, tools & testing	UMG written in C++, editor commandlets and Python editor scripts that generate assets, automation tests, debug overlays and console commands

EXPERIENCE

UEFN Developer · Zeritress, freelance with creator partners — 2023 – present

- Shipped **12 Fortnite islands** with 6 creator partners: **855M+ minutes played**, a 44,962 all-time player peak, 482 published versions.
- Built Verse systems: persistent economies and save state, shops and item granting, per-player state tested with 50 players, prestige and ranked progression.
- Ran live-ops at about two updates a week on flagship islands, including PlayStation vs Xbox (552M minutes played).

AI Programmer & Technical Animator · Moshpit Studios

- NPC behaviour for a multiplayer virtual concert with AI Controllers and behavior trees; technical animation for crowd and player characters; tested NPCs against player powers.

C++ Contributor · Malakhra: Blood & Mayhem, 6-person team, UE 5.8 — 2026

- Built menu-hub systems in C++ (fighter cages, vanity wall, options, clickable navigation) and an editor-only camera-anchor visualizer for level artists.

SELECTED PROJECTS

Field Correspondent · solo, UE 5.8 C++ — 2026

- LLM-driven NPC: the model proposes actions through tool calls, C++ checks them against a whitelist and the NavMesh, and a behavior tree with six custom C++ nodes carries them out. Interrupts in about 0.2 s.
- Hybrid retrieval with citations (0.044 ms median search), AI Perception into timed memories. **120 automation tests** and 50 end-to-end checks passing in a packaged build.

Ashford Hollow · solo, UE 5.8 C++, 2–4 player networked RPG — 2026

- About 18,600 lines of C++: GAS combat (melee combo with input buffering, predicted ranged attack, area slam, dash strike), server-authoritative hits and loot, a quest-giver NPC.
- Enemy AI: one AI Controller with Perception, a behavior tree per enemy type, EQS positioning, and attack tokens for a pack that flanks.
- Server-side lag compensation that **cut ghost hits from 30% to 2% at 300 ms**, measured with a scripted test client.

GASShooter port to UE 5.8 — 2026

- Ported tranek's GAS shooter sample from 5.3 to 5.8 (removed montage replication fields, instancing policy changes, prediction keys) and fixed an inverted ensure that blocked server-side target-data callbacks.

Third-person action RPG · final year project, UE5, carried forward in UE 5.8 C++ — 2026

- Melee combat with elemental abilities, enemy AI, XP and a three-branch skill tree across two levels.
- In UE 5.8: a soulslike boss fight against a Stone Golem, a new player character retargeted at runtime onto the existing combat, and inventory, crafting, loot and a key-locked gate in C++, wired to the Blueprints by reflection.
- A Dark Souls-style C++ HUD (damage trails, a boss bar with the flurry total, enemy bars in sight), the skill tree restyled at runtime, **10 automation tests**.